GALOIS-INVARIANT CLOSED SUBSETS OF THE BASE CHANGE DESCEND TO THE GROUND SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted statement and its proof are self-contained with respect to the pinned file: the proof cites no result of the project, so this task carries no external source references and no reference stubs.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-varieties/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup, of the native theorem declarations and mathematical macros, and of the project's own no-op reference/slogan/history display environments. The native class line and the xr-hyper externaldocument block are omitted because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The Galois-theoretic facts used at proof lines 921-927 (finiteness of the stage carrying a given element, and the invariants of a finite Galois extension) appear among the standard prerequisites rather than as context spans, because the pinned file proves neither.

Allowed standard prerequisites (editorial).

- Basic Galois theory: invariants of a finite Galois extension and the orbit polynomial. Used to see that the coefficients of P(T) = prod_{sigma in G}(T - sigma(x)) lie in A; the construction of the polynomial, the passage to a finite stage and the radical argument are all retained.
- Radical ideals and the correspondence between closed subsets and radical ideals. The closed subset T-bar is encoded by its radical ideal I-bar inside the base change of A to k-bar; the proof shows that I-bar equals the radical of the ideal generated by I in that base change, and reads off T = V(I).

Lemma 0.1. *Let k be a field. Let $\overline{k}/k$ be a (possibly infinite) Galois extension. Let X be a scheme over k. Let $\overline{T} \subset X_{\overline{k}}$ have the following properties*

(1) *$\overline{T}$ is a closed subset of $X_{\overline{k}}$,*
(2) *for every $\sigma \in Gal(\overline{k}/k)$ we have $\sigma(\overline{T}) = \overline{T}$.*

Then there exists a closed subset $T \subset X$ whose inverse image in $X_{\overline{k}}$ is $\overline{T}$.

Proof. This lemma immediately reduces to the case where $X = \mathrm{Spec}(A)$ is affine. In this case, let $\overline{I} \subset A \otimes_k \overline{k}$ be the radical ideal corresponding to $\overline{T}$. Assumption (2) implies that $\sigma(\overline{I}) = \overline{I}$ for all $\sigma \in \mathrm{Gal}(\overline{k}/k)$. Pick $x \in \overline{I}$. There exists a finite Galois extension k'/k contained in $\overline{k}$ such that $x \in A \otimes_k k'$. Set $G = \mathrm{Gal}(k'/k)$. Set

$$P(T) = \prod\nolimits_{\sigma \in G} (T - \sigma(x)) \in (A \otimes_k k')[T]$$

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It is clear that $P(T)$ is monic and is actually an element of $(A \otimes_k k')^G[T] = A[T]$ (by basic Galois theory). Moreover, if we write $P(T) = T^d + a_1 T^{d-1} + \ldots + a_d$ the we see that $a_i \in I := A \cap \overline{I}$. Combining $P(x) = 0$ and $a_i \in I$ we find $x^d = -a_1 x^{d-1} - \ldots - a_d \in I(A \otimes_k \overline{k})$. Thus x is contained in the radical of $I(A \otimes_k \overline{k})$. Hence $\overline{I}$ is the radical of $I(A \otimes_k \overline{k})$ and setting $T = V(I)$ is a solution. $\square$